



ECE375

Large Number Arithmetic

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Goal of this Lab

- Build arithmetic functions.
- Manipulate large numbers in Assembly.
- Properly use subroutines.
- Use memory windows to observe data.

Large Number Arithmetic

- Assembly use 8-bit instruction sets.
- Figure out how to build arithmetic operations on numbers that are larger than 8 bits
- Operands are provided in skeleton code.
- Simulator-based Lab (Use ATmega128)

16bit Adder



	\$A2	\$FF
	\$F4	\$77
\$01	\$97	\$76

Program Memory to Data Memory

Program Memory

\$A2	\$FF	\$0200
\$F4	\$77	\$0201
\$00	\$00	\$0202

Data Memory

\$D1	\$0100
\$D2	\$0101
\$D3	\$0102
\$D4	\$0103

Program Memory to Data Memory



Program Memory

\$A2	\$FF	\$0200
\$F4	\$77	\$0201
\$00	\$00	\$0202

Data Memory

\$FF	\$0100
\$A2	\$0101
\$77	\$0102
\$F4	\$0103

Data Memory to Register

Program Memory

\$A2	\$FF	\$0200
\$F4	\$77	\$0201
\$00	\$00	\$0202

Data Memory

\$FF	\$0100
\$A2	\$0101
\$77	\$0102
\$F4	\$0103

Register

	\$A2	\$FF
	\$F4	\$77
+		
	\$01	\$97
		\$76

Operations

ADD +

SUB -

MUL ×

Register to Data Memory

Program Memory

\$A2	\$FF	\$0200
\$F4	\$77	\$0201
\$00	\$00	\$0202

Data Memory

\$FF	\$0100
\$A2	\$0101
\$77	\$0102
\$F4	\$0103
\$76	\$0104
\$97	\$0105
\$01	\$0106

	\$A2	\$FF
	\$F4	\$77
+	\$01	\$97
	\$76	

16bit Adder

+		\$A2	\$FF
		\$F4	\$77
	\$01	\$97	\$76

+	\$0101 : \$0100
	\$0103 : \$0102
	\$0106 : \$0105 : \$0104

16bit Adder

FUNCTION:

ldi XL, \$00

ldi XH, \$01

ldi YL, \$02

ldi YH, \$01

ldi ZL, \$04

ldi ZH, \$01

\$A2	\$FF	\$0101 : \$0100
\$F4	\$77	\$0103 : \$0102
\$00	\$00	\$0105 : \$0104
\$00	\$00	\$0107 : \$0106



16bit Adder

FUNCTION:

ldi XL, \$00

ldi XH, \$01

ldi YL, \$02

ldi YH, \$01

ldi ZL, \$04

ldi ZH, \$01

\$A2	\$FF	\$0101 : \$0100
\$F4	\$77	\$0103 : \$0102
\$00	\$00	\$0105 : \$0104
\$00	\$00	\$0107 : \$0106

X

Y

Z

16bit Adder

FUNCTION:

ldi XL, \$00

ldi XH, \$01

ldi YL, \$02

ldi YH, \$01

ldi ZL, \$04

ldi ZH, \$01

\$A2	\$FF	\$0101 : \$0100
\$F4	\$77	\$0103 : \$0102
\$00	\$00	\$0105 : \$0104
\$00	\$00	\$0107 : \$0106



16bit Adder

FUNCTION:

ldi XL, \$00

ldi XH, \$01

ldi YL, \$02

ldi YH, \$01

ldi ZL, \$04

ldi ZH, \$01

\$A2	\$FF	\$0101 : \$0100
\$F4	\$77	\$0103 : \$0102
\$00	\$00	\$0105 : \$0104
\$00	\$00	\$0107 : \$0106



16bit Adder

FUNCTION:

ldi XL, \$00

ldi XH, \$01

ldi YL, \$02

ldi YH, \$01

ldi ZL, \$04

ldi ZH, \$01

\$A2	\$FF	\$0101 : \$0100
\$F4	\$77	\$0103 : \$0102
\$00	\$00	\$0105 : \$0104
\$00	\$00	\$0107 : \$0106

X

Y

Z

16bit Adder

```
ld    R16,X+
ld    R17,Y+
add   R17,R16
st    Z+,R17
ld    R16,X
ld    R17,Y
adc   R17,R16
st    Z+,R17
brcc  EXIT
st    Z,XH
EXIT:
      ret
```

\$A2	\$FF	\$0101 : \$0100
\$F4	\$77	\$0103 : \$0102
\$00	\$00	\$0105 : \$0104
\$00	\$00	\$0107 : \$0106

\$16	R16
\$17	R17
C=0	SREG

16bit Adder

ld R16, X+

ld R17, Y+

add R17, R16

st Z+, R17

ld R16, X

ld R17, Y

adc R17, R16

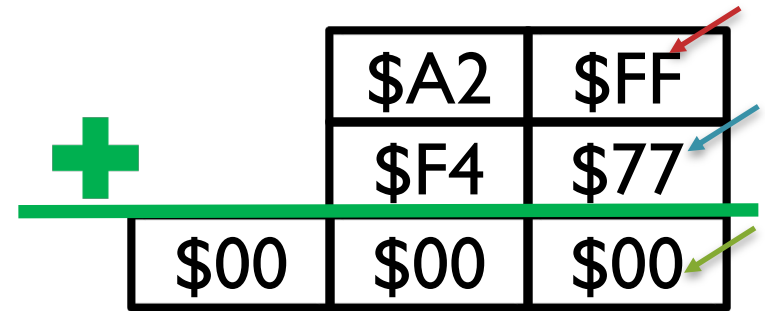
st Z+, R17

brcc EXIT

st Z, XH

EXIT:

ret



\$A2	\$FF	\$0101 : \$0100
\$F4	\$77	\$0103 : \$0102
\$00	\$00	\$0105 : \$0104
\$00	\$00	\$0107 : \$0106

\$FF	R16
\$17	R17
C=0	SREG

16bit Adder

ld R16, X+

ld R17, Y+

add R17, R16

st Z+, R17

ld R16, X

ld R17, Y

adc R17, R16

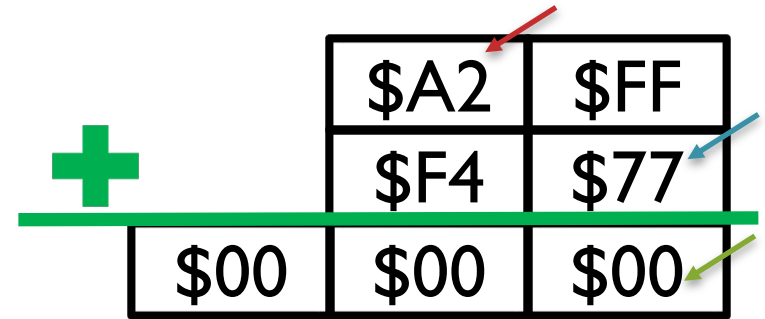
st Z+, R17

brcc EXIT

st Z, XH

EXIT:

ret



\$A2	\$FF	\$0101 : \$0100
\$F4	\$77	\$0103 : \$0102
\$00	\$00	\$0105 : \$0104
\$00	\$00	\$0107 : \$0106

\$FF	R16
\$17	R17
C=0	SREG

16bit Adder

ld R16, X+

ld R17, Y+

add R17, R16

st Z+, R17

ld R16, X

ld R17, Y

adc R17, R16

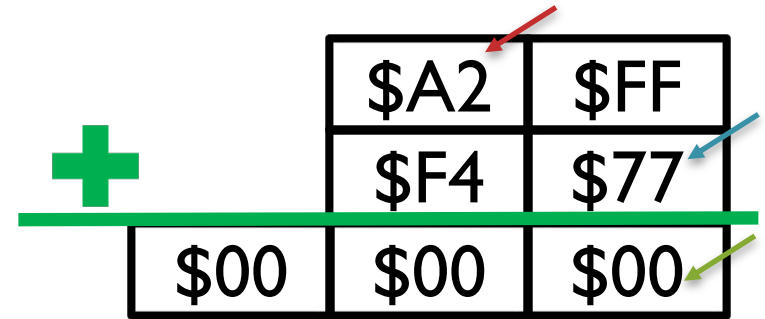
st Z+, R17

brcc EXIT

st Z, XH

EXIT:

ret



\$A2	\$FF	\$0101 : \$0100
\$F4	\$77	\$0103 : \$0102
\$00	\$00	\$0105 : \$0104
\$00	\$00	\$0107 : \$0106

\$FF	R16
\$77	R17
C=0	SREG

16bit Adder

ld R16, X+

ld R17, Y+

add R17, R16

st Z+, R17

ld R16, X

ld R17, Y

adc R17, R16

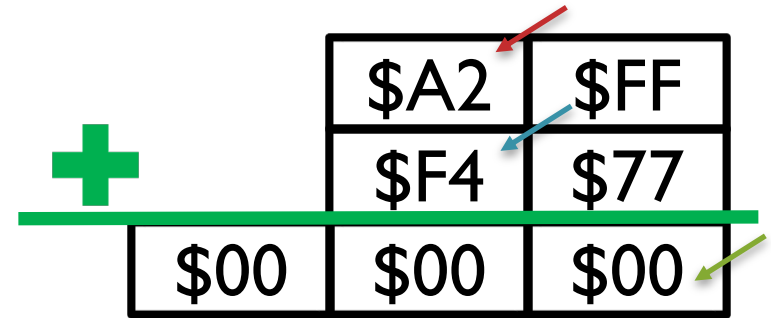
st Z+, R17

brcc EXIT

st Z, XH

EXIT:

ret



\$A2	\$FF	\$0101 : \$0100
\$F4	\$77	\$0103 : \$0102
\$00	\$00	\$0105 : \$0104
\$00	\$00	\$0107 : \$0106

\$FF	R16
\$77	R17
C=0	SREG

16bit Adder

ld R16, X+

ld R17, Y+

add R17, R16

st Z+, R17

ld R16, X

ld R17, Y

adc R17, R16

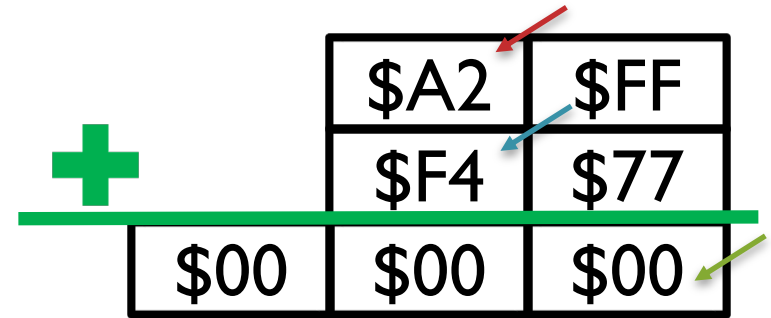
st Z+, R17

brcc EXIT

st Z, XH

EXIT:

ret



\$A2	\$FF	\$0101 : \$0100
\$F4	\$77	\$0103 : \$0102
\$00	\$00	\$0105 : \$0104
\$00	\$00	\$0107 : \$0106

\$FF	R16
\$77	R17
C=0	SREG

16bit Adder

ld R16, X+

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add R17, R16

st Z+, R17

ld R16, X

ld R17, Y

adc R17, R16

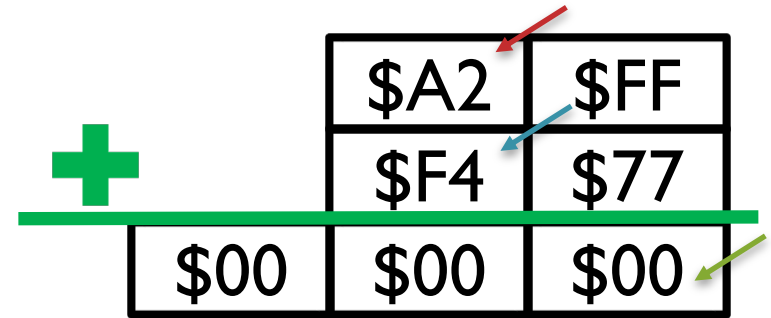
st Z+, R17

brcc EXIT

st Z, XH

EXIT:

ret



\$A2	\$FF	\$0101 : \$0100
\$F4	\$77	\$0103 : \$0102
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\$00	\$00	\$0107 : \$0106

\$FF	R16
\$76	R17
C=1	SREG

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ld R17, Y+

add R17, R16

st Z+, R17

ld R16, X

ld R17, Y

adc R17, R16

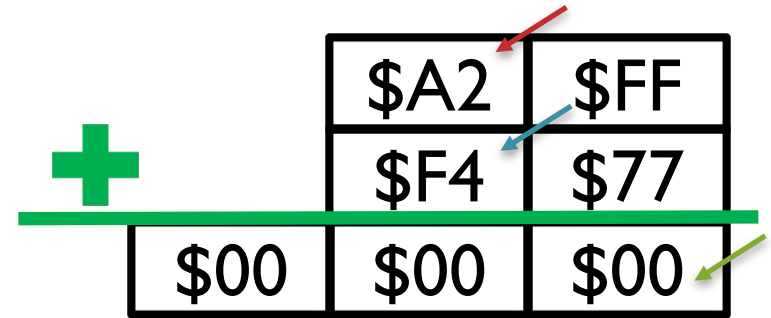
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EXIT:

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\$A2	\$FF	\$0101 : \$0100
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add R17, R16

st Z+, R17

ld R16, X

ld R17, Y

adc R17, R16

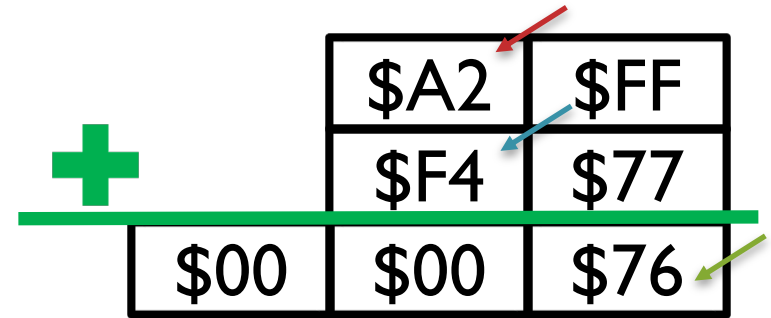
st Z+, R17

brcc EXIT

st Z, XH

EXIT:

ret



\$A2	\$FF	\$0101 : \$0100
\$F4	\$77	\$0103 : \$0102
\$00	\$76	\$0105 : \$0104
\$00	\$00	\$0107 : \$0106

\$FF	R16
\$76	R17
C=1	SREG

16bit Adder

ld R16, X+

ld R17, Y+

add R17, R16

st Z+, R17

ld R16, X

ld R17, Y

adc R17, R16

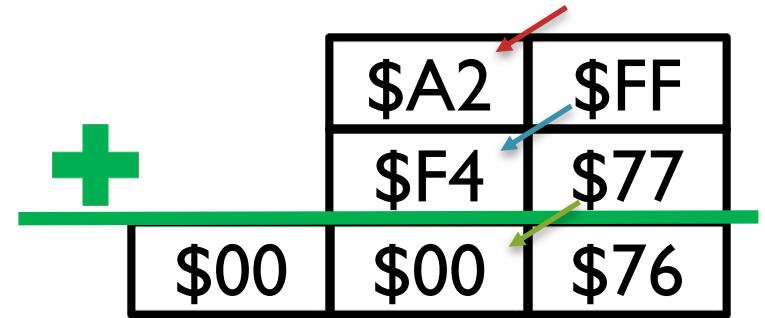
st Z+, R17

brcc EXIT

st Z, XH

EXIT:

ret



\$A2	\$FF	\$0101 : \$0100
\$F4	\$77	\$0103 : \$0102
\$00	\$76	\$0105 : \$0104
\$00	\$00	\$0107 : \$0106

\$FF	R16
\$76	R17
C=1	SREG

16bit Adder

ld R16, X+

ld R17, Y+

add R17, R16

st Z+, R17

ld R16, X

ld R17, Y

adc R17, R16

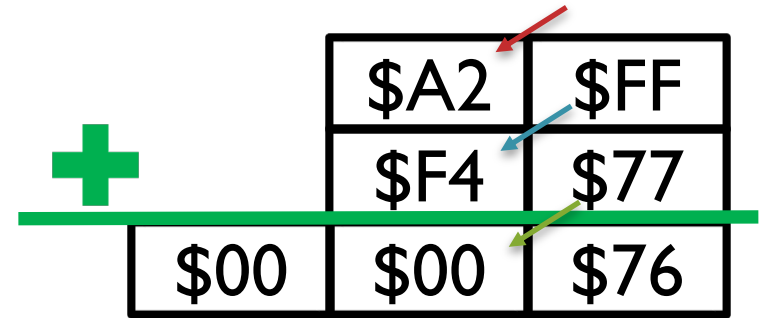
st Z+, R17

brcc EXIT

st Z, XH

EXIT:

ret



\$A2	\$FF	\$0101 : \$0100
\$F4	\$77	\$0103 : \$0102
\$00	\$76	\$0105 : \$0104
\$00	\$00	\$0107 : \$0106

\$A2	R16
\$76	R17
C=1	SREG

16bit Adder

ld R16, X+

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add R17, R16

st Z+, R17

ld R16, X

ld R17, Y

adc R17, R16

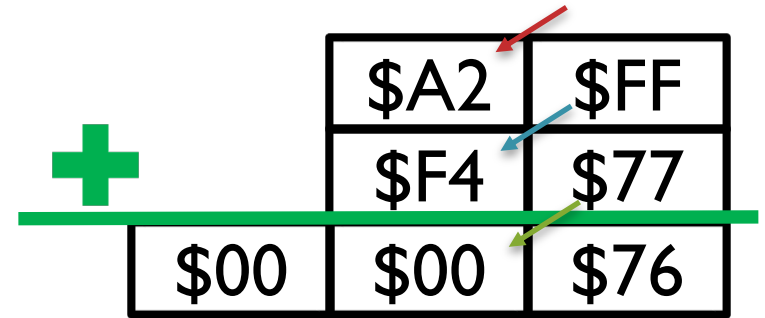
st Z+, R17

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EXIT:

ret



\$A2	\$FF	\$0101 : \$0100
\$F4	\$77	\$0103 : \$0102
\$00	\$76	\$0105 : \$0104
\$00	\$00	\$0107 : \$0106

\$A2	R16
\$F4	R17
C=1	SREG

16bit Adder

ld R16, X+

ld R17, Y+

add R17, R16

st Z+, R17

ld R16, X

ld R17, Y

adc R17, R16

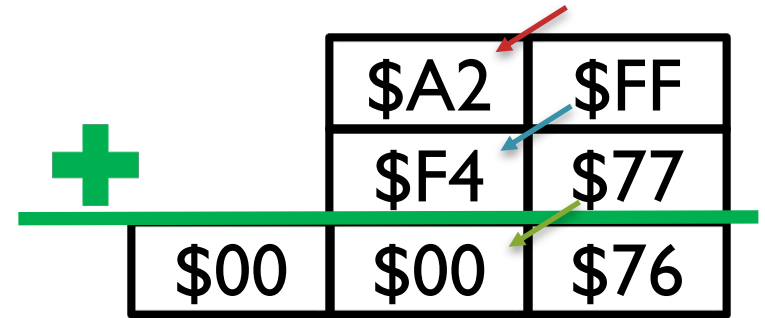
st Z+, R17

brcc EXIT

st Z, XH

EXIT:

ret



\$A2	\$FF	\$0101 : \$0100
\$F4	\$77	\$0103 : \$0102
\$00	\$76	\$0105 : \$0104
\$00	\$00	\$0107 : \$0106

\$A2	R16
\$97	R17
C=1	SREG

16bit Adder

ld R16, X+

ld R17, Y+

add R17, R16

st Z+, R17

ld R16, X

ld R17, Y

adc R17, R16

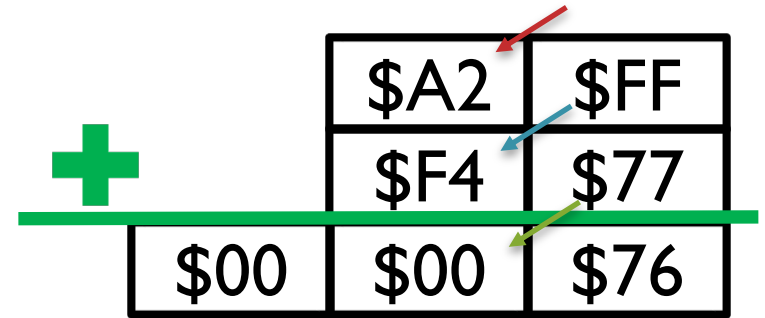
st Z+, R17

brcc EXIT

st Z, XH

EXIT:

ret



\$A2	\$FF	\$0101 : \$0100
\$F4	\$77	\$0103 : \$0102
\$00	\$76	\$0105 : \$0104
\$00	\$00	\$0107 : \$0106

\$A2	R16
\$97	R17
C=1	SREG

16bit Adder

ld R16, X+

ld R17, Y+

add R17, R16

st Z+, R17

ld R16, X

ld R17, Y

adc R17, R16

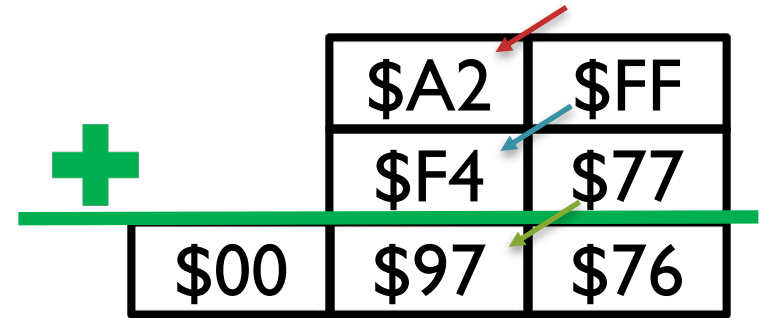
st Z+, R17

brcc EXIT

st Z, XH

EXIT:

ret



\$A2	\$FF	\$0101 : \$0100
\$F4	\$77	\$0103 : \$0102
\$97	\$76	\$0105 : \$0104
\$00	\$00	\$0107 : \$0106

\$A2	R16
\$97	R17
C=1	SREG

16bit Adder

ld R16, X+

ld R17, Y+

add R17, R16

st Z+, R17

ld R16, X

ld R17, Y

adc R17, R16

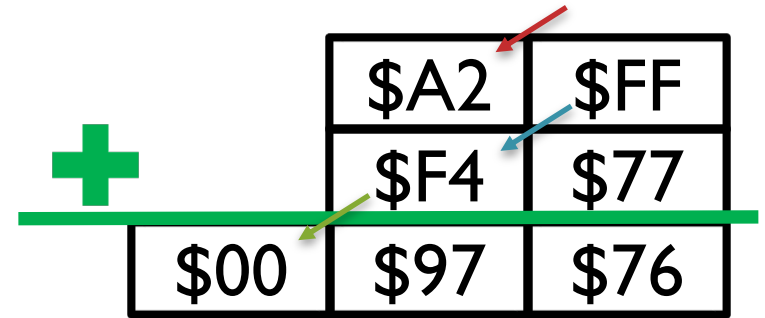
st Z+, R17

brcc EXIT

st Z, XH

EXIT:

ret



\$A2	\$FF	\$0101 : \$0100
\$F4	\$77	\$0103 : \$0102
\$97	\$76	\$0105 : \$0104
\$00	\$00	\$0107 : \$0106

\$A2	R16
\$97	R17
C=1	SREG

16bit Adder

ld R16, X+

ld R17, Y+

add R17, R16

st Z+, R17

ld R16, X

ld R17, Y

adc R17, R16

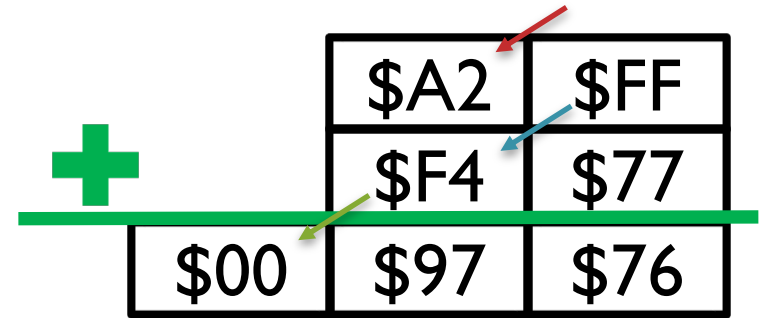
st Z+, R17

brcc EXIT

st Z, XH

EXIT:

ret



\$A2	\$FF	\$0101 : \$0100
\$F4	\$77	\$0103 : \$0102
\$97	\$76	\$0105 : \$0104
\$00	\$00	\$0107 : \$0106

\$FF	R16
\$76	R17
C=1	SREG

16bit Adder

ld R16, X+

ld R17, Y+

add R17, R16

st Z+, R17

ld R16, X

ld R17, Y

adc R17, R16

st Z+, R17

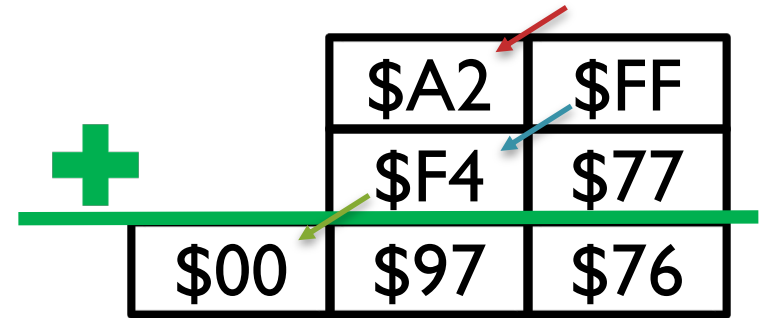
brcc EXIT

st Z, XH

EXIT:

ret

ldi R18, \$01
st. Z, R18



\$A2	\$FF	\$0101 : \$0100
\$F4	\$77	\$0103 : \$0102
\$97	\$76	\$0105 : \$0104
\$00	\$00	\$0107 : \$0106

\$FF	R16
\$76	R17
C=1	SREG

16bit Adder

ld R16, X+

ld R17, Y+

add R17, R16

st Z+, R17

ld R16, X

ld R17, Y

adc R17, R16

st Z+, R17

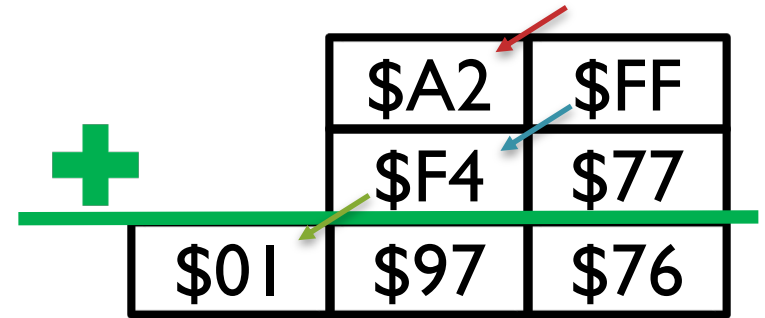
brcc EXIT

st Z, XH

EXIT:

ret

ldi R18, \$01
st. Z, R18



\$A2	\$FF	\$0101 : \$0100
\$F4	\$77	\$0103 : \$0102
\$97	\$76	\$0105 : \$0104
\$00	\$01	\$0107 : \$0106

\$FF	R16
\$76	R17
C=1	SREG

Check-off Lists

- Correctly defined operands.
- Correct data manipulation for operands.
- Correct results for each arithmetic function.

Announcements

- We don't use AVR board for this lab.

Questions?

